

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Indicative content: Description of the types of radiation Alpha is a helium nucleus. Beta is a fast moving electron. Gamma is an em wave. Properties of the types of radiation Alpha is highly ionising but weakly penetrating, absorbed by paper. Beta is quite ionising and quite penetrating, it is blocked by thin aluminium. Gamma is highly penetrating, blocked by thick lead but not very ionising. Consequences for storage Need suitable shielding / lead to prevent gamma radiation escaping. Needs to be in leak proof / earthquake proof containers to prevent contamination of water supplies. Needs to be stored safely for a really long time as some waste contains isotopes with very long half-lives. Should be well away from people / buried deep underground. 5–6 marks Comprehensive coverage of all 3 areas: description, properties and waste storage. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				3–4 marks Comprehensive coverage of 2 areas: description, properties and waste storage or limited description of all 3. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks Comprehensive coverage of 1 area: description, properties and waste storage or limited description of 1 or 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)			$^{235}_{92}\text{U} (1) \rightarrow ^{231}_{90}\text{Th} + ^4_2\alpha (1)$ $^{231}_{90}\text{Th} \rightarrow ^{231}_{91}\text{Pa} (1) + ^0_{-1}\beta (1)$	1	1		4		
				Question 4 total	8	2	0	10	0	0